

Figure 17.3 Difference between http and https

Secure Sockets Layers

The most common Cryptographic protocol is Secure Sockets Layers (SSL). SSL is a hybrid encryption protocol for securing transactions over the Internet. The SSL standard was developed by Netscape in collaboration with MasterCard, Bank of America, MCI and Silicon Graphics. It is based on a public key cryptography process to ensure the security of data transmission over the internet. Its principle is to establish a secure communication channel (encrypted) between a client and a server after an authentication step.

The SSL system acts as an additional layer, to ensure the security of data, located between the application layer and the transport layer in TCP. For example, a user using an internet browser to connect to an SSL secured E-Commerce site will send encrypted data without any more necessary manipulations. Secure Sockets Layers (SSL) was renamed as Transport Layer Security (TLS) in 2001. But still it is popularly known under the name SSL. TLS differs from SSL in the generation of symmetric keys.

Today, all browsers in the market support SSL, and most of the secure communications are proceeded through this protocol. SSL works completely hidden for the user, who does not have to intervene in the protocol. The only thing the user has to do is make sure the URL starts with https:// instead of http:// where the "s" obviously means secured. It is also preceded by a green padlock.

3D Secure

"3-D Secure is a secure payment protocol on the Internet. It was developed by Visa to increase the level of transaction security, and it has been adapted by MasterCard. It gives a better authentication of the holder of the payment card, during purchases made on websites. The basic concept of this (XML-based) protocol is to link the financial authorization process with an online authentication system. This authentication model comprise 3 domains (hence the name 3D) which are:

1. The Acquirer Domain
2. The Issuer Domain
3. The interoperability Domain

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